

Dataset Analysis for Decision-Making

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 **sicor_municipal_co... .csv**
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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To analyze the dataset and address your request, I will first load the dataset (`sicor_municipal_contracts.csv`) and examine its structure and key characteristics. This will help identify the variables available for analysis and ensure that subsequent queries can be answered effectively.

Step 1: Load and Inspect the Dataset

I'll start by loading the dataset using Python's `pandas` library and inspecting its contents to understand the structure, data types, and any missing values.

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(data.head())
```

Step 2: Key Characteristics of the Dataset

From the output of the above code, I will extract the following:

- Number of Rows and Columns** : Total records and variables.
- Column Names and Data Types** : Understanding what each column represents and its data type (e.g., numeric, categorical, date).
- Missing Values** : Identifying columns with missing or null values.
- Summary Statistics** : For numeric columns, provide descriptive statistics.

```
python
1 # Summary statistics for numeric columns
2 print("\nSummary Statistics for Numeric Columns:")
3 print(data.describe())
4
5 # Check for missing values
6 print("\nMissing Values per Column:")
7 print(data.isnull().sum())
```

Step 3: Address a Specific Analytical Question

Once you provide a specific analytical question, I will use the dataset to answer it. For example, potential questions could include:

- What is the total value of contracts awarded in a specific year?
- Which municipalities have the highest number of contracts?
- What is the average contract value by category?



Please provide the specific question you'd like addressed, and I will write the corresponding Python code and analysis to answer it.

Let me know how you'd like to proceed!

How can the total contract value be reliably estimated based on financial institution profiles, resource sources, and program codes? Define three approaches to address this issue and select the best one.



To reliably estimate the total contract value based on financial institution profiles, resource sources, and program codes, we need to analyze the relationships between these variables and the contract values in the dataset. Below, I outline three potential approaches to address this issue, followed by a recommendation for the best approach.

Approach 1: Group-Based Aggregation

This approach involves grouping the data by relevant categorical variables (e.g., financial institution profiles, resource sources, and program codes) and calculating the average or median contract value for each group. These aggregated values can then serve as estimates for future contracts with similar characteristics.

Steps:

1. Group the dataset by the relevant variables (`financial_institution` , `resource_source` , `program_code`).
2. Calculate the mean or median contract value for each group.
3. Use these group-level statistics as the estimated contract value for new contracts.

Pros:

- Simple and interpretable.
- Requires minimal assumptions about the data distribution.

Cons:

- May not capture complex interactions between variables.
- Less accurate if the dataset contains outliers or high variability within groups.

Approach 2: Regression Modeling

This approach uses statistical or machine learning regression models to predict contract values based on the provided variables. The model would treat `contract_value` as the dependent variable and use `financial_institution` , `resource_source` , and `program_code` (encoded as numerical or categorical features) as predictors.

Steps:

1. Preprocess the data:
 - Encode categorical variables (e.g., one-hot encoding for `financial_institution` , `resource_source` , and `program_code`).
 - Handle missing values if any exist.
2. Split the data into training and testing sets.
3. Train a regression model (e.g., linear regression, decision tree, or random forest) on the training set.
4. Evaluate the model's performance on the test set using metrics like Mean Absolute Error (MAE) or Root Mean Squared Error (RMSE).

Pros:

- Can capture non-linear relationships and interactions between variables.
- Provides a flexible framework for incorporating additional predictors.

Cons:

- Requires more computational effort.
- May overfit if the dataset is small or noisy.

Approach 3: Clustering-Based Estimation

This approach clusters contracts based on their characteristics (e.g., financial institution profiles, resource sources, and



program codes). The average contract value of each cluster is then used as an estimate for contracts with similar features.

Steps:

1. Normalize the numerical features (if any) and encode categorical variables.
2. Apply a clustering algorithm (e.g., K-Means or hierarchical clustering) to group contracts into clusters.
3. Calculate the average contract value for each cluster.
4. Assign new contracts to the nearest cluster and use the cluster's average value as the estimate.

Pros:

- Captures natural groupings in the data.
- Useful when relationships between predictors and outcomes are not well understood.

Cons:

- Difficult to interpret the meaning of clusters.
- Sensitive to the choice of clustering algorithm and parameters.

Recommendation: Approach 2 (Regression Modeling)

Among the three approaches, **regression modeling** is the most robust and flexible for estimating total contract values. It allows us to leverage the full range of predictors and their interactions, while also providing a clear framework for evaluating model performance. Additionally, regression models can be extended to include more complex algorithms (e.g., gradient boosting) if needed.

Below is the Python code to implement Approach 2:

Python Code for Regression Modeling

```
python
1 import pandas as pd
2 from sklearn.model_selection import train_test_split
3 from sklearn.ensemble import RandomForestRegressor
4 from sklearn.metrics import mean_absolute_error
5
6 # Load the dataset
7 file_path = 'sior_municipal_contracts.csv'
8 data = pd.read_csv(file_path)
9
10 # Select relevant features and target variable
11 features = ['financial_institution', 'resource_source', 'program_code']
12 target = 'contract_value'
13
14 # Encode categorical variables
15 data_encoded = pd.get_dummies(data[features])
16
17 # Handle missing values (if any)
18 data_encoded = data_encoded.fillna(0)
19
20 # Define features and target
21 X = data_encoded
22 y = data[target]
23
24 # Split the data into training and testing sets
25 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
26
27 # Train a Random Forest Regressor
28 model = RandomForestRegressor(random_state=42)
29 model.fit(X_train, y_train)
30
31 # Make predictions on the test set
32 y_pred = model.predict(X_test)
33
34 # Evaluate the model
35 mae = mean_absolute_error(y_test, y_pred)
36 print(f"Mean Absolute Error: {mae:.2f}")
37
38 # Example: Estimate contract value for a new contract
39 new_contract = pd.DataFrame({
40     'financial_institution': ['BankA'],
41     'resource_source': ['Source1'],
42     'program_code': ['Code123']
43 })
44 new_contract_encoded = pd.get_dummies(new_contract).reindex(columns=X.columns, fill_value=0)
45 predicted_value = model.predict(new_contract_encoded)
```



```
46 print(f"Estimated Contract Value: {predicted_value[0]:.2f}")
```

Explanation of the Code

1. **Data Encoding** : Categorical variables are converted into numerical format using one-hot encoding.
2. **Model Training** : A Random Forest Regressor is trained on the encoded features and target variable (`contract_value`).
3. **Evaluation** : The model's performance is evaluated using Mean Absolute Error (MAE).
4. **Prediction** : The trained model is used to estimate the contract value for a new contract based on its characteristics.

Conclusion

The regression modeling approach provides a reliable and flexible method for estimating contract values. By leveraging machine learning techniques, we can capture complex relationships in the data and produce accurate predictions. If you have specific requirements or constraints, feel free to share them, and I can refine the solution further.

